International Cybersecurity and Digital Forensics Academy (ICDFA)

Basic Computer Skills for Digital Forensics

Case Studies: Reconstructing Chrome Web History

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EXECUTIVE SUMMARY

This investigation focused on reconstructing web browsing activity from a Google Chrome History database as part of a digital forensics exercise. The objective was to examine browser artefacts, establish a timeline of recorded web activity, identify search terms and download records, and correlate related browsing events.

The investigation involved examining the SQLite-based Chrome History database, validating the integrity of the acquired evidence, analysing relevant database tables, and exporting query results for further review. The analysis identified 57 URL records, 74 visit records, four search terms, and one download record. The recorded browsing activity covered the period from 19 April 2022 at 13:54:18 UTC to 15:02:56 UTC.

The investigation also identified cryptocurrency-related browsing activity, including visits to blockchain and cryptocurrency exchange websites, as well as a recorded download named proof_of_payment.png. These artefacts provided leads for further examination and correlation. However, browser history alone does not establish that a transaction was completed, that a particular person owned an account, or that the user performed every recorded action.

Evidence integrity was assessed by comparing the original and working copies using SHA-256 hashes and a byte-for-byte comparison. Both database copies produced the same hash, and the SQLite integrity check returned ok. SQL queries were used to extract

and organise the relevant artefacts, with results retained in CSV and text formats to support reproducibility and reporting.

Overall, the investigation demonstrated how browser history artefacts can support the reconstruction of browsing activity, event sequencing, and the identification of potentially relevant digital evidence.

PURPOSE OF THE ASSIGNMENT

The purpose of this assignment was to apply digital forensic techniques to a Google Chrome History database to reconstruct browsing activity and demonstrate how browser artefacts can support a structured digital investigation.

The exercise aimed to develop practical skills in evidence preservation, integrity verification, SQLite database examination, SQL-based artefact extraction, timeline reconstruction, and the interpretation of browser activity. It also emphasised the importance of documenting investigative procedures and distinguishing between findings supported by digital evidence and conclusions that require additional corroboration.

OBJECTIVES OF THE ASSIGNMENT

The objectives of the assignment were to:

1. Acquire and preserve the Chrome History database for forensic examination.
2. Verify the integrity of the original evidence and working copy using cryptographic hashing and byte-level comparison.
3. Examine the database schema to identify tables and fields relevant to browsing activity.
4. Extract and analyse recorded URLs, visit events, search terms, and download information.
5. Reconstruct a chronological timeline of browser activity using the available timestamps.
6. Identify potentially relevant browsing patterns and relationships between recorded web visits.
7. Export query results into structured formats for documentation, analysis, and reporting.

8. Maintain an evidence integrity log and investigative notes to support transparency and reproducibility.
9. Interpret findings objectively while acknowledging the limitations of browser history as a source of evidence.

METHODOLOGY OF THE ASSIGNMENT

The investigation followed a structured digital forensics workflow comprising evidence preparation, integrity verification, database examination, artefact extraction, timeline reconstruction, and documentation.

1. Evidence acquisition and preparation

The Chrome History database was placed in an evidence directory and a separate working copy was maintained for analysis. The original evidence was preserved to minimise the risk of accidental modification during examination.

2. Evidence integrity verification

SHA-256 hashes were calculated for the original and working database files. A byte-for-byte comparison was also performed to confirm that the files were identical. The SQLite PRAGMA integrity_check command was executed against the working copy to assess the database's internal consistency. The results were recorded in the evidence integrity log.

3. Database schema examination

The database structure was examined to identify tables relevant to browser activity, including urls, visits, keyword_search_terms, downloads, downloads_url_chains, and meta. The schema examination helped determine which fields could be used to extract browsing events, search terms, download records, and associated metadata.

4. SQL-based artefact extraction

SQL queries were executed against the working database to obtain record counts, browsing timestamps, search terms, download details, and potentially relevant URL records. Separate queries were used to generate a comprehensive visit timeline. The extracted results were saved as text and CSV files for subsequent analysis and reporting.

5. Timeline reconstruction and correlation

Recorded visit timestamps were converted to UTC and arranged chronologically to establish the sequence of browsing activity. Related URL records and page titles were

examined to identify potentially meaningful relationships between visits, including cryptocurrency-related browsing and the recorded image download.

6. Interpretation and documentation

The results were interpreted in the context of the available evidence. Findings were documented in investigation notes, while the SQL scripts and query outputs were retained to support reproducibility. Conclusions were limited to what the available browser artefacts could reasonably establish, with attribution and transaction completion treated as matters requiring independent corroboration.

TOOLS AND RESOURCES USED

The following tools and resources were used during the investigation to preserve evidence integrity, examine the Chrome History database, extract relevant artefacts, reconstruct browsing activity, and document the findings.

S/N	Tool/Resource	Purpose
1	Kali Linux	Provided the forensic working environment for evidence management, database analysis, and investigation documentation.
2	SQLite3	Used to inspect the Chrome History database, examine its schema, execute SQL queries, and perform database integrity checks.
3	Structured Query Language (SQL)	Used to extract URL records, visit history, search terms, download records, metadata, and chronological browsing activity.
4	SHA-256	Used to generate cryptographic hashes for verifying the integrity of the original evidence and working copy.
5	Linux command-line utilities	Utilities including sha256sum, cmp, stat, file, find, and tee were used for hashing, file comparison, evidence identification, inventory, and logging.
6	Chrome History database	Served as the primary digital evidence source for reconstructing recorded browser activity.

7	CSV and text files	Used to store query results, browsing timelines, database schema details, and investigative findings for review and reporting.
8	Git	Used to track changes and version-control the investigation artefacts.
9	GitHub	Used to host and organise the investigation files, SQL scripts, evidence database, query results, and supporting documentation in a repository.

These tools and resources supported a structured and reproducible investigation, from evidence integrity verification through artefact extraction, timeline reconstruction, and documentation.

Introduction and Investigation Objective

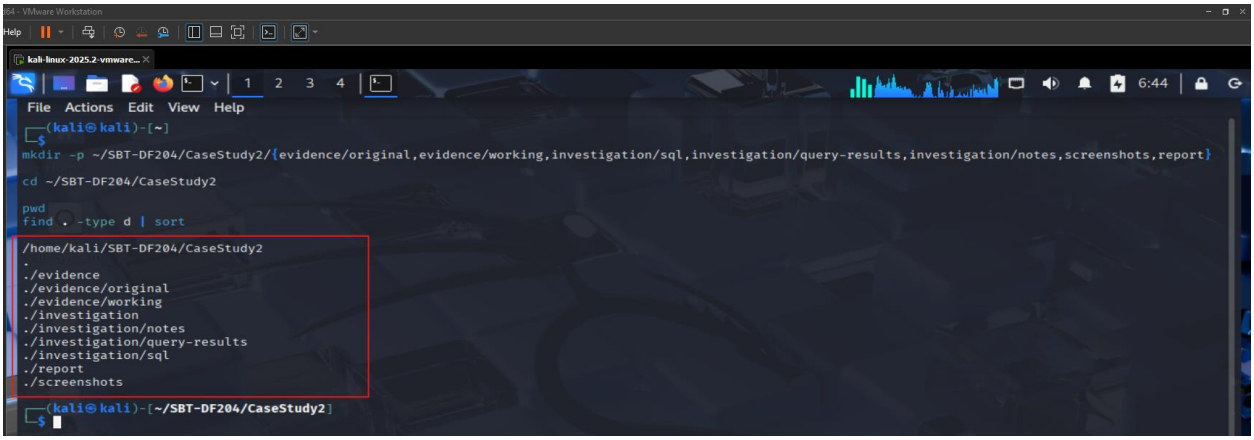
This case study examines a Google Chrome browser History database to reconstruct a sequence of recorded web-navigation activities using digital forensic techniques. The investigation aims to identify relevant browser records, establish the chronology of activity, examine navigation relationships, and correlate available browsing and download artefacts where supported by the evidence. The analysis will distinguish direct observations from investigative interpretations and will avoid attributing browser activity conclusively to a specific individual without independent corroboration.

The examination will be conducted using Kali Linux and SQLite analysis tools. The original evidence will be preserved, and analysis will be performed on a separate working copy. Findings will be documented through SQL queries, labelled screenshots, a chronological timeline expressed in Coordinated Universal Time (UTC), and an evidence log.

Creation of the Case Directory Structure: Investigation Environment and Workspace Preparation

The investigation workspace was created in the Kali Linux home directory at /home/kali/SBT-DF204/CaseStudy2. A structured directory hierarchy was established to separate evidence preservation from examination activities and report preparation. The evidence/original directory is designated for the preserved original database, while

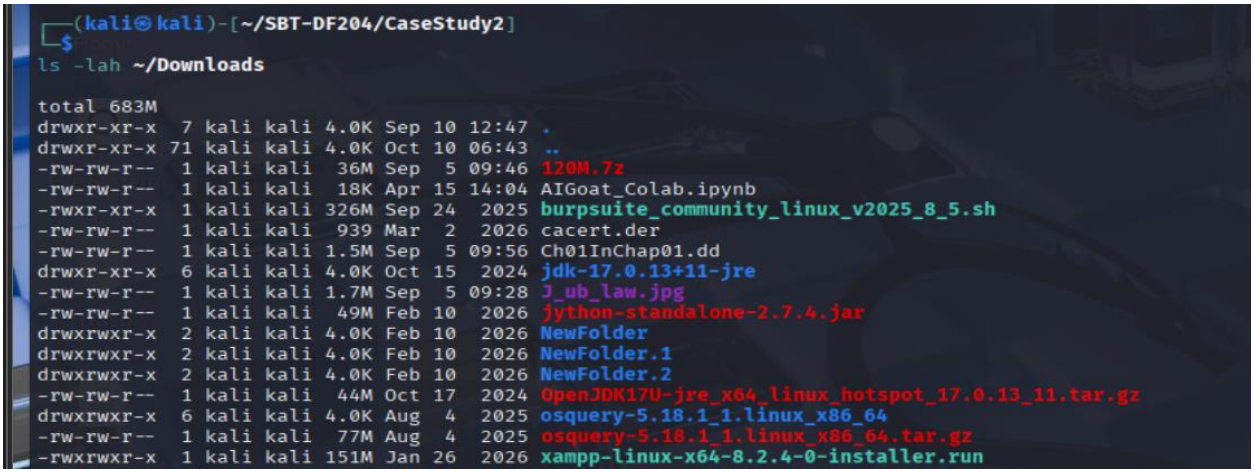
evidence/working is intended for the examination copy. The investigation/sql directory will contain the SQL queries used during analysis, and investigation/query-results will hold relevant query outputs. The investigation/notes directory will contain examination notes, while screenshots and report will store the visual evidence and final written report, respectively.



This separation supports an organised and reproducible workflow by keeping the original evidence distinct from files created during examination. At this stage, the directory structure has been created successfully, but the Chrome History database has not yet been located or acquired. Therefore, no conclusions about the contents of the browser history have been drawn.

Identification of the Evidence Location

The Kali Linux Downloads directory was inspected using the ls -lah ~/Downloads command to identify the supplied Chrome History database or a possible archive containing it. The command displayed the files and directories currently present in the Downloads folder, including their sizes, permissions, and modification dates. However, the resulting listing did not visibly identify a file named History or an obvious archive containing the required database.



Consequently, the evidence source remains to be located. The supplied database must be obtained from the course study materials before acquisition and examination can proceed. No unrelated file will be substituted for the assessment evidence, and the personal Chrome profile will not be used as an alternative source.

Verification of Available Examination Utilities

The availability of the required command-line utilities was checked using the which command. The results confirmed that sqlite3 is available at /usr/bin/sqlite3, sha256sum at /usr/bin/sha256sum, and file at /usr/bin/file. These utilities will support database examination, cryptographic integrity verification, and preliminary file-type identification, respectively.

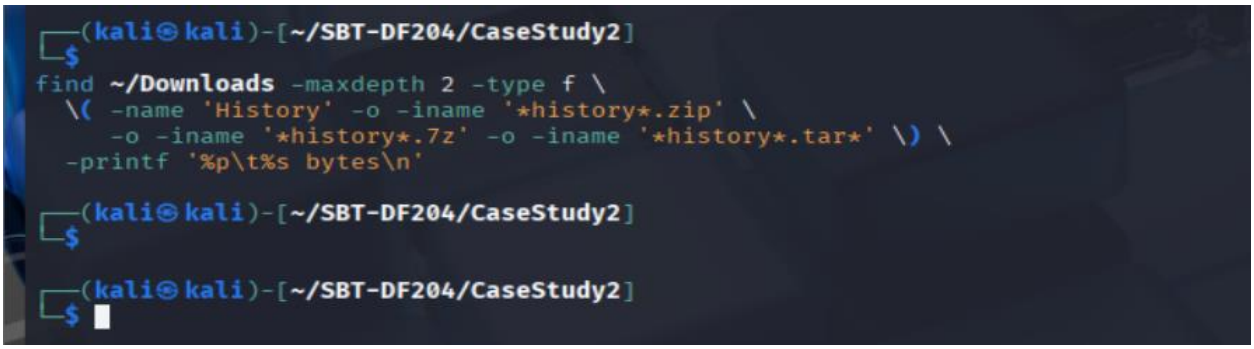


The SQLite command-line utility was also checked using sqlite3 --version. The installed version was reported as SQLite 3.46.1, released on 13 August 2024. This confirms that a suitable SQLite command-line examination utility is available in the environment. The successful version check establishes tool availability only; it does not establish the

integrity or validity of the evidence database, which will be assessed after the file is acquired.

Initial Evidence Search and Search Limitation

An initial search was conducted within the Kali Linux Downloads directory to locate the Chrome History database or a recognised archive containing the assessment evidence. The search used the find utility with a maximum directory depth of two and filename patterns targeting a file named History and archives with names containing history and extensions such as .zip, .7z, and .tar. The command returned no results, indicating that no files matching the specified criteria were found within the searched location and depth.



This result does not establish that the evidence is unavailable; the database may be stored elsewhere, contained in an archive with a different filename, or still accessible only through the course platform. Further searches will be conducted before the acquisition stage is considered complete. No substitute evidence will be introduced into the investigation.

Extended Search for Browser History Artefacts

Following the unsuccessful search of the Downloads directory, an extended filename search was performed across the /home/kali directory. The search identified two files named History: /home/kali/.BurpSuite/pre-wired-browser/Default/History, with a reported size of 163,840 bytes, and /home/kali/.config/chromium/Default/History, with a reported size of 360,448 bytes.

```
┌──(kali㉿kali)-[~/SBT-DF204/CaseStudy2]
└─$ find /home/kali -type f \
  \( -iname 'History' \
     -o -iname 'History.db' \
     -o -iname '*history*.zip' \
     -o -iname '*history*.7z' \
     -o -iname '*history*.tar' \
     -o -iname '*history*.tar.gz' \
     -o -iname '*history*.tgz' \
     -o -iname '*chrome*.zip' \
     -o -iname '*chrome*.7z' \) \
  -printf '%p\t%s bytes\n' 2>/dev/null
/home/kali/.BurpSuite/pre-wired-browser/Default/History 163840 bytes
/home/kali/.config/chromium/Default/History     360448 bytes

┌──(kali㉿kali)-[~/SBT-DF204/CaseStudy2]
└─$
```

These results establish that browser history databases exist within the local Kali environment. However, their locations indicate that they are associated with the Burp Suite pre-wired browser and the local Chromium profile, respectively. Neither file has been established as the artefact supplied for the assessment. Consequently, both files have been excluded from the current examination pending confirmation of the authorised evidence source. This precaution prevents unrelated browser activity from being incorrectly presented as evidence for the case study.

Review of Recent Downloads

A review of the 40 most recently modified regular files within the Downloads directory and its first two directory levels was conducted to identify potential assessment evidence. The output included files such as Ch01InChap01.dd and 120M.7z; however, neither filename establishes a connection to the Chrome Web History case study. No file was identified from this listing as the required Chrome History database or its associated evidence archive.

```
┌──(kali㉿kali)-[~/SBT-DF204/CaseStudy2]
└─$ find ~/Downloads -maxdepth 2 -type f \
  -printf '%TY-%Tm-%Td %TH:%TM\t%s bytes\t%p\n' \
  | sort -r | head -40
2026-09-05 09:56        1474560 bytes   /home/kali/Downloads/Ch01InChap01.dd
2026-09-05 09:46        36720470 bytes  /home/kali/Downloads/120M.7z
2026-09-05 09:28        1692150 bytes   /home/kali/Downloads/J_ub_law.jpg
2026-04-15 14:04        17567 bytes     /home/kali/Downloads/AIGoat_Colab.ipynb
2026-03-02 16:48        939 bytes       /home/kali/Downloads/cacert.der
2026-02-10 13:33        50453449 bytes  /home/kali/Downloads/jython-standalone-2.7.4.jar
2026-01-26 09:52        157746191 bytes /home/kali/Downloads/xampp-linux-x64-8.2.4-0-installer.run
2025-09-24 08:53        341424340 bytes /home/kali/Downloads/burpsuite_community_linux_v2025_8_5.sh
2025-08-04 07:03        80367951 bytes  /home/kali/Downloads/osquery-5.18.1_1.linux_x86_64.tar.gz
2024-10-17 17:12        46082619 bytes  /home/kali/Downloads/OpenJDK17U-jre_x64_linux_hotspot_17.0.13_11.tar.gz
2024-10-15 19:26        2400 bytes      /home/kali/Downloads/jdk-17.0.13+11-jre/NOTICE
2024-10-15 19:26        1631 bytes      /home/kali/Downloads/jdk-17.0.13+11-jre/release

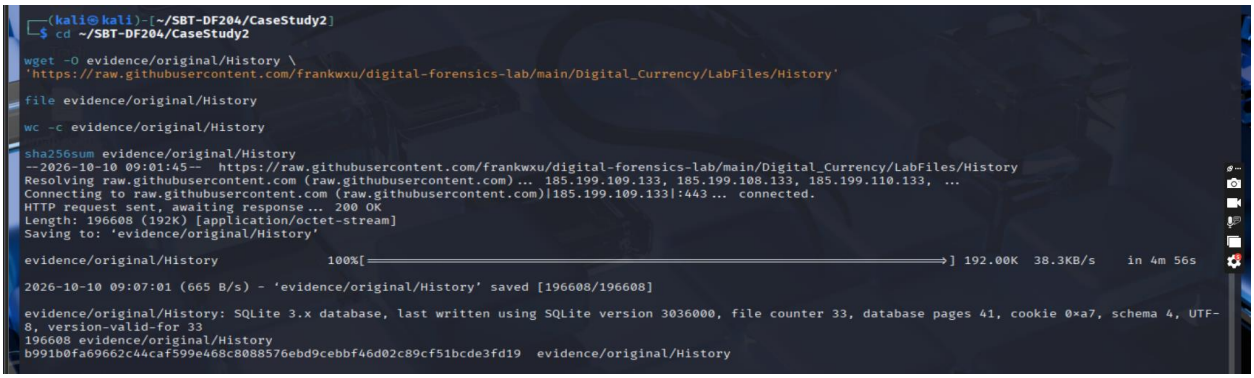
┌──(kali㉿kali)-[~/SBT-DF204/CaseStudy2]
└─$
```

The search was therefore treated as preliminary, and the relevance of any candidate file remains unverified. The assessment brief and its specified evidence source must be

consulted before any artefact is acquired or examined. No conclusions about browser activity have been drawn at this stage.

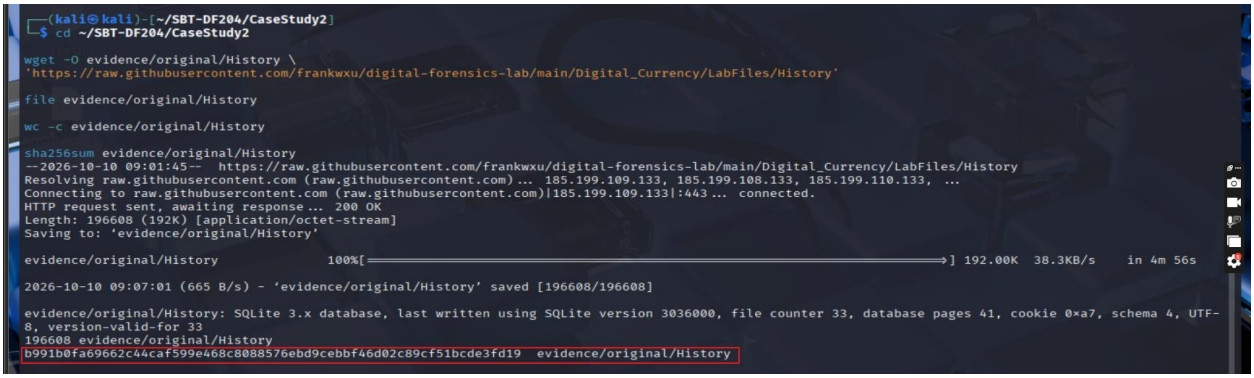
Evidence Acquisition and Initial Verification of the Chrome History Database

The Chrome History database was acquired from the Digital Forensics Lab repository hosted on GitHub, using the wget utility in the Kali Linux examination environment. The file was downloaded from the Digital_Currency/LabFiles/History path and saved as evidence/original/History within the Case Study 2 workspace. The acquisition command returned an HTTP 200 OK response, indicating that the remote server successfully served the requested resource. The download completed successfully, with a reported file size of 196,608 bytes (192 KiB), consistent with the size stated in the available laboratory instructions.



File Identification and Integrity Record

Following acquisition, the file utility identified the downloaded artefact as an SQLite 3.x database. The output also reported database metadata, including 41 database pages, UTF-8 encoding, and a last-written SQLite version of 3.36.0. The wc -c command confirmed the exact file size as 196,608 bytes. A SHA-256 hash was then calculated to establish a reference value for subsequent integrity checks. The recorded hash was b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19.

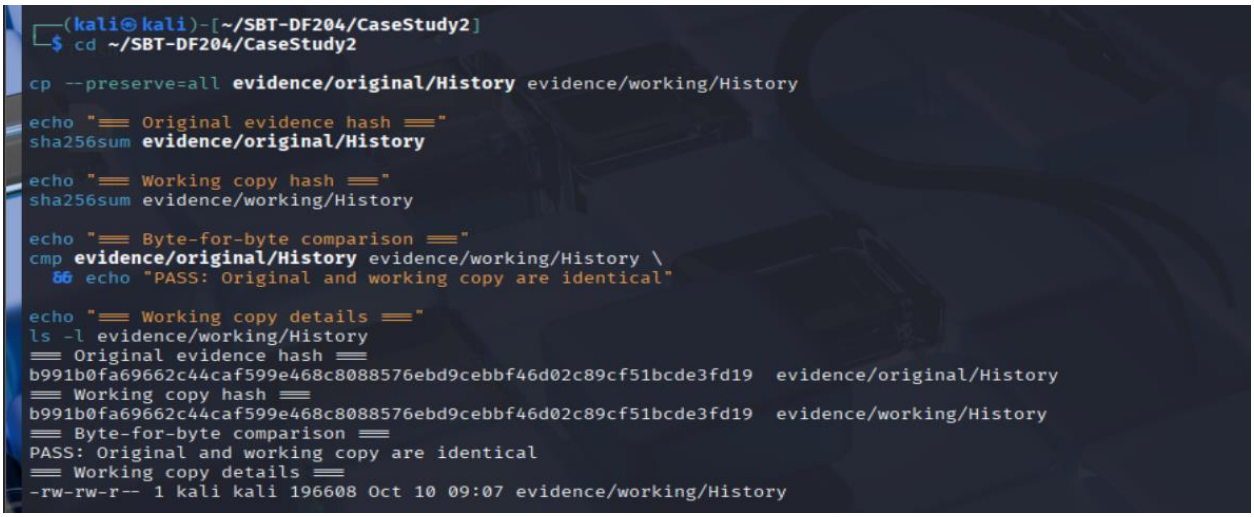


Acquisition and initial verification of the Chrome History database. The terminal output documents the successful download, reported transfer size, SQLite file identification, exact byte count, and calculated SHA-256 hash.

These checks establish the downloaded file's size, identify its apparent format, and provide a hash value against which subsequent copies can be compared. However, because no independently supplied hash for the source artefact was available at this stage, the calculated hash alone does not establish that the downloaded file is identical to a separately held original. The source and relevance of this repository artefact must also be assessed against the Case Study 2 assignment instructions before findings are attributed to the case.

Creation and Verification of the Working Copy

To preserve the acquired evidence and reduce the risk of modifying it during examination, a separate working copy of the Chrome History database was created using the Linux cp --preserve=all command. The acquired file was retained at evidence/original/History, while the copy intended for analysis was stored at evidence/working/History. The command preserved the original file's available attributes during copying.



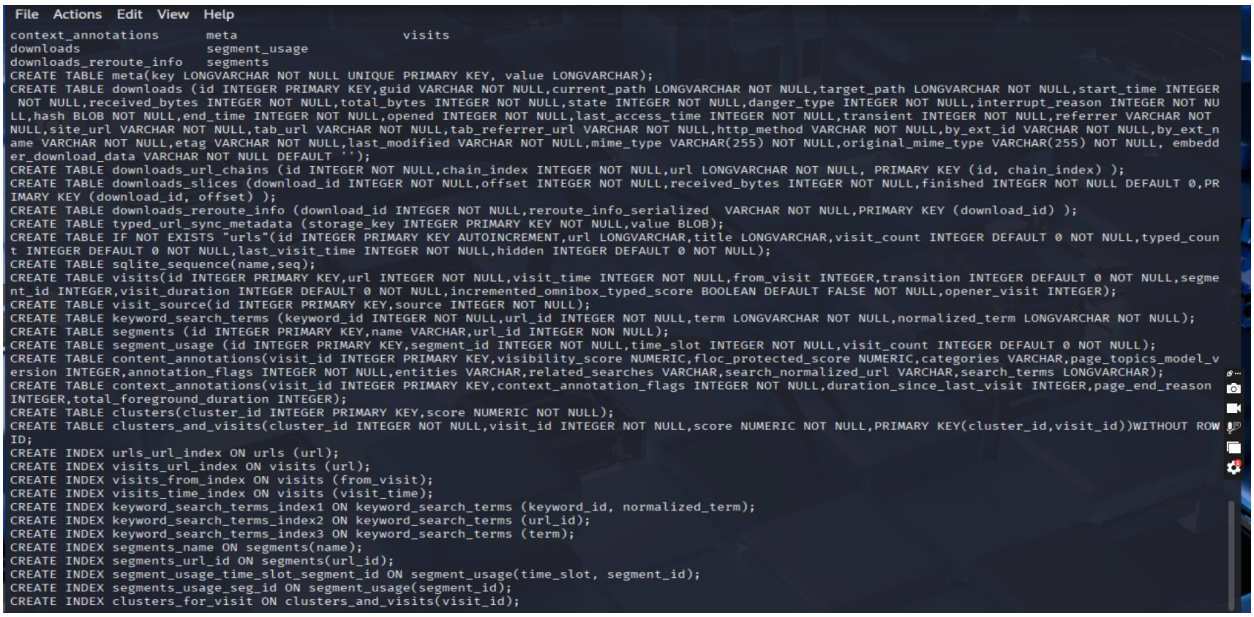
The integrity of the working copy was verified by calculating the SHA-256 hash of both files and comparing them with the cmp utility. Both files produced the hash b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19. The byte-for-byte comparison returned PASS, confirming that the working copy was identical to the acquired file at the time of verification. The working copy's reported size was 196,608 bytes.

The original file will be retained as the reference copy, and subsequent database examination will be performed on the working copy. Hash verification will be repeated if needed to check that the evidence files remain unchanged during the investigation.

Examination of the Chrome History Database Schema

The SQLite schema of the working copy was examined using the SQLite command-line utility. The database contains tables associated with browsing history, navigation events, search terms, downloads, and browser-generated annotations. The urls table contains URL strings, page titles, visit counts, typed counts, and last-visit timestamps. The visits table records individual visit events and includes fields for visit timestamps, referring visits, navigation transitions, and visit duration. These tables can be correlated through the URL identifier to reconstruct recorded browsing activity and establish relationships between navigation events.





The database also contains the keyword_search_terms table, which may preserve search terms associated with browser search activity, and the downloads and downloads_url_chains tables, which can provide information about recorded downloads and their URL chains. Additional tables, including visit_source, content_annotations, and context_annotations, may offer supporting metadata. Their presence does not, by itself, establish that the corresponding tables contain relevant records.

The schema was exported to investigation/notes/database-schema.txt to maintain a record of the database structure used to guide the examination. The next stage will enumerate the records in the principal tables and inspect the available timestamps and relationships before developing the detailed browsing timeline. No conclusions about the user's intentions or the meaning of any particular browsing activity are drawn from the schema alone.

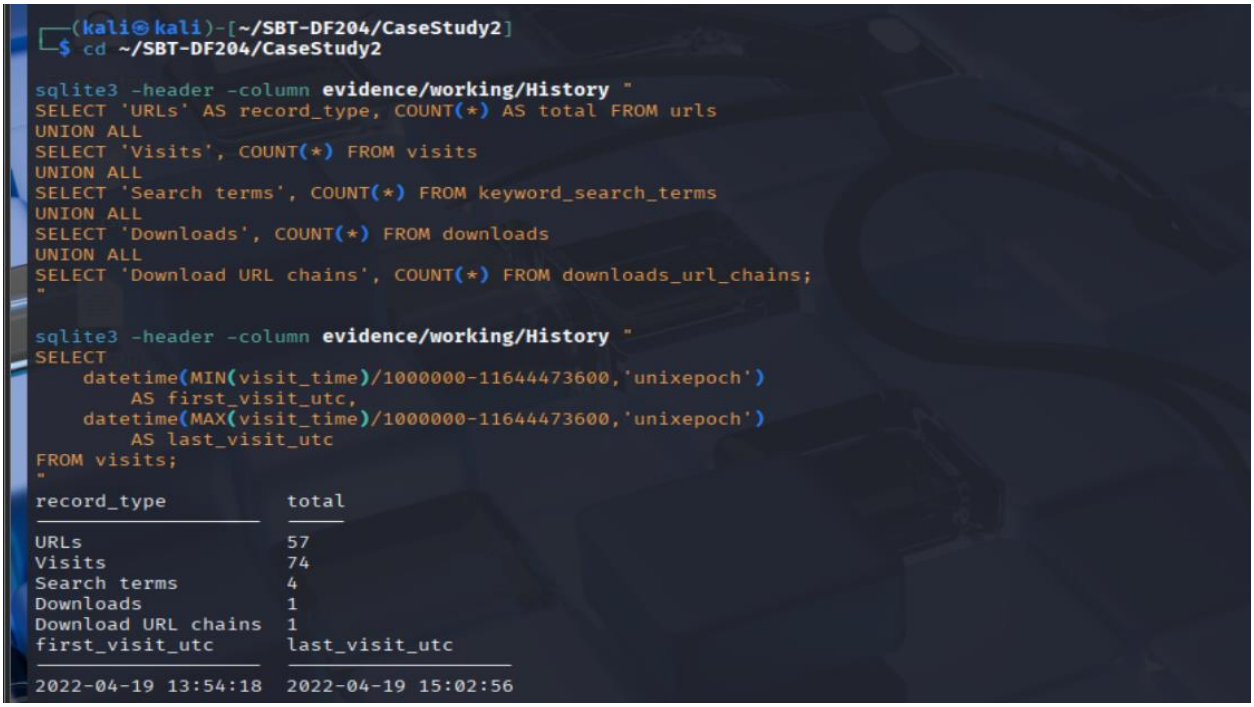
Core tables identified in the Chrome History database

Table	Key fields	Investigative relevance
urls	id, url, title, visit_count, typed_count, last_visit_time	Stores URL records, page titles, visit counts, typed counts, and last-visit timestamps.
visits	id, url, visit_time, from_visit, transition, visit_duration, opener_visit	Records individual visits and supports chronological reconstruction and navigation relationship analysis.
keyword_search_terms	keyword_id, url_id, term, normalized_term	May contain search terms associated with browser search activity.
downloads	id, target_path, start_time, end_time, received_bytes, total_bytes, state, mime_type, referrer	Records download metadata, including paths, timing, size, state, and referrer information.
downloads_url_chains	id, chain_index, url	Records URL chains associated with downloads and may help identify their source or redirects.
meta	key, value	Stores database metadata that may help contextualise the database.

visit_source	id, source	Stores source classifications for visit records where available.
content_annotations	visit_id, categories, entities, search_terms	May provide additional page or search-related annotations, depending on the records present.

Database Record Inventory and Temporal Coverage

Following schema identification, read-only SQL queries were executed against the working copy to establish the number of records in the principal Chrome History tables and determine the temporal coverage of the recorded browsing visits. The database contained 57 records in the urls table and 74 records in the visits table. The keyword_search_terms table contained four records, while the downloads and downloads_url_chains tables each contained one record.



The earliest recorded visit timestamp was 19 April 2022 at 13:54:18 UTC, and the latest was 19 April 2022 at 15:02:56 UTC. The observed visit records therefore span approximately 68 minutes and 38 seconds. This range describes the earliest and latest timestamps present in the examined visits table; it does not, by itself, establish the full

period during which the browser was used or prove that the database contains a complete browsing history.

The record inventory provides an initial overview of the available evidence and informs the next stage of examination. Subsequent queries will retrieve individual URLs, page titles, visit timestamps, search terms, and download metadata. These records will be correlated where appropriate to reconstruct a timeline while distinguishing direct database

Initial Chrome History record counts and visit-time coverage

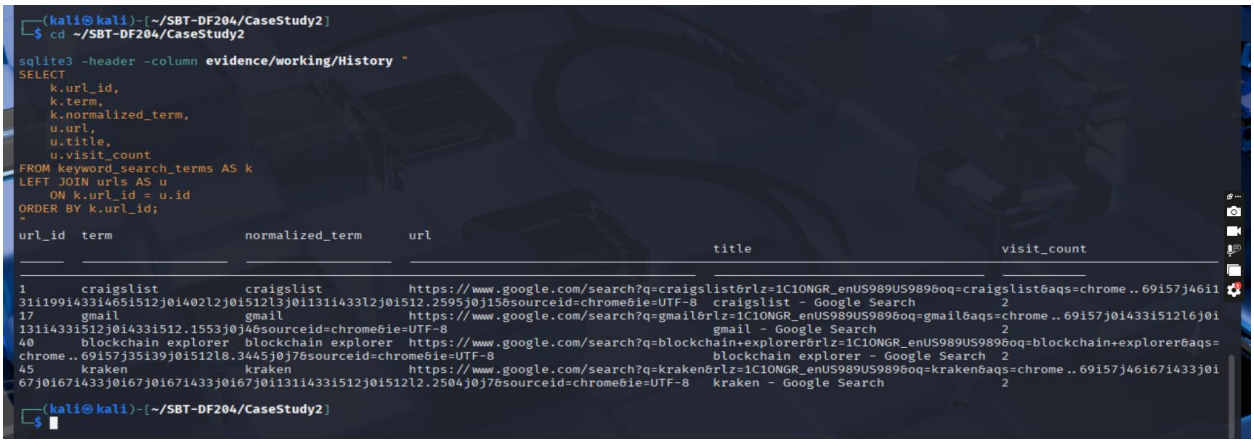
Record category	Number of records
URL records (urls)	57
Visit records (visits)	74
Search-term records (keyword_search_terms)	4
Download records (downloads)	1
Download URL-chain records (downloads_url_chains)	1
Earliest recorded visit (UTC)	19 April 2022, 13:54:18
Latest recorded visit (UTC)	19 April 2022, 15:02:56
Observed timestamp span	1 hour, 8 minutes, 38 seconds

Examination of Recorded Search Terms

A read-only SQL query was used to retrieve records from the keyword_search_terms table and associate them with their corresponding URL entries in the urls table. The query returned four search-term records: “craigslist,” “gmail,” “blockchain explorer,” and “kraken.” Each record was associated with a Google Search URL whose page title reflected the corresponding term. The stored visit_count for each associated URL was two.

The results establish that these four terms are represented in the examined browser database and are linked to Google Search URL records. The stored visit counts relate to the corresponding URL entries and should not be interpreted as proof of two distinct searches by a particular person. Likewise, the presence of terms associated with

cryptocurrency services or blockchain exploration does not independently establish cryptocurrency ownership, exchange activity, a financial transaction, or unlawful conduct. Further examination of individual visit timestamps, related URLs, and download records is required to assess the context and sequence of the recorded browsing activity.



Search terms recovered from the Chrome History database

No.	Recorded search term	Associated page title	Stored URL visit count
1	craigslist	craigslist - Google Search	2
2	gmail	gmail - Google Search	2
3	blockchain explorer	blockchain explorer - Google Search	2
4	kraken	kraken - Google Search	2

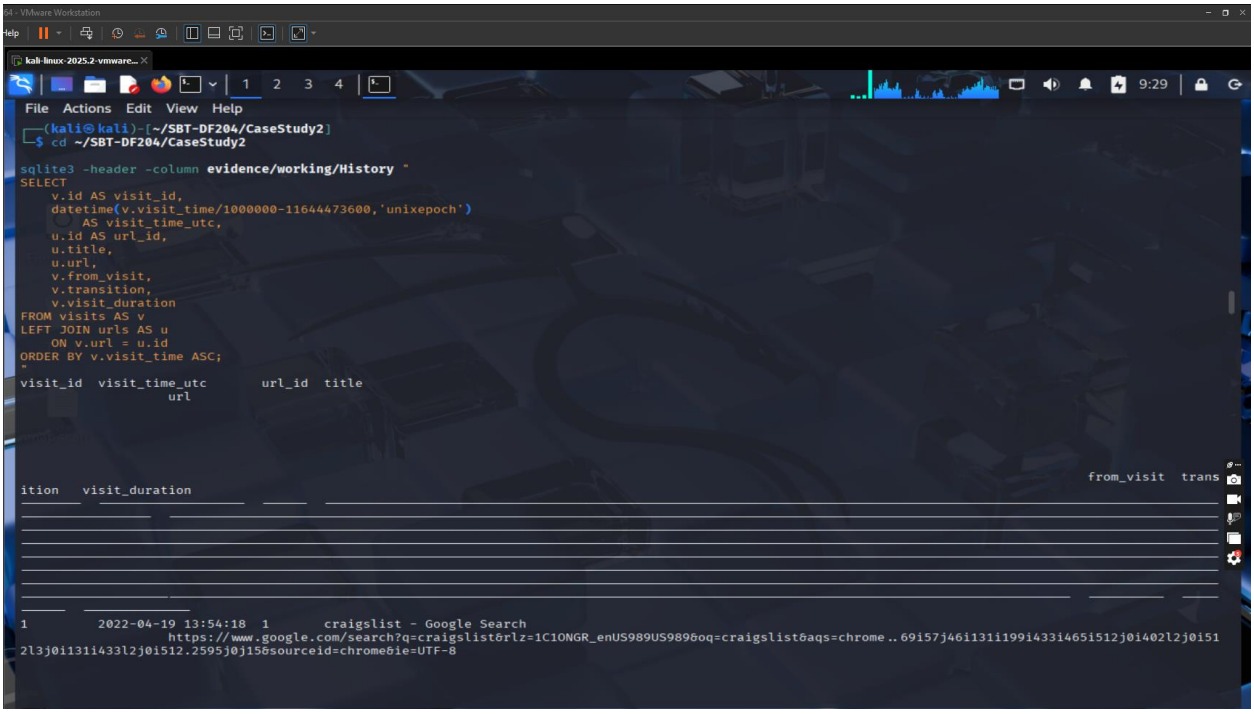
Preliminary Reconstruction of Browsing Activity

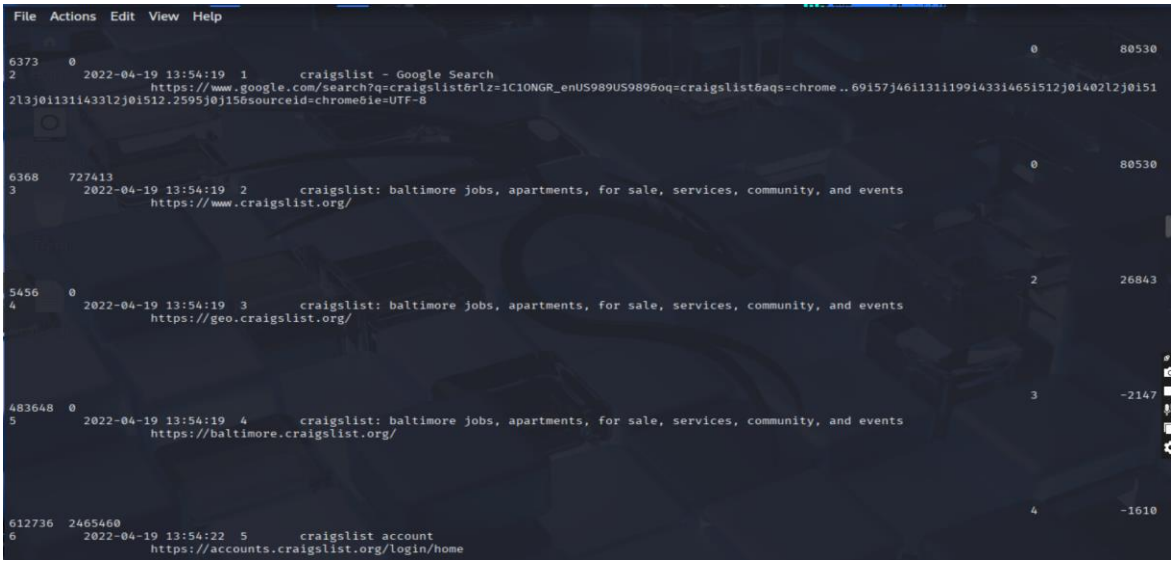
The 74 records in the visits table were ordered by their converted UTC timestamps to establish a preliminary browsing sequence. The earliest recorded activity involved a Google search for “craigslist” at 13:54:18 UTC, followed by Craigslist homepage, account, and posting workflow pages. Between 13:55:05 and 13:58:27 UTC, the database records successive pages associated with selecting a posting type and category, editing a

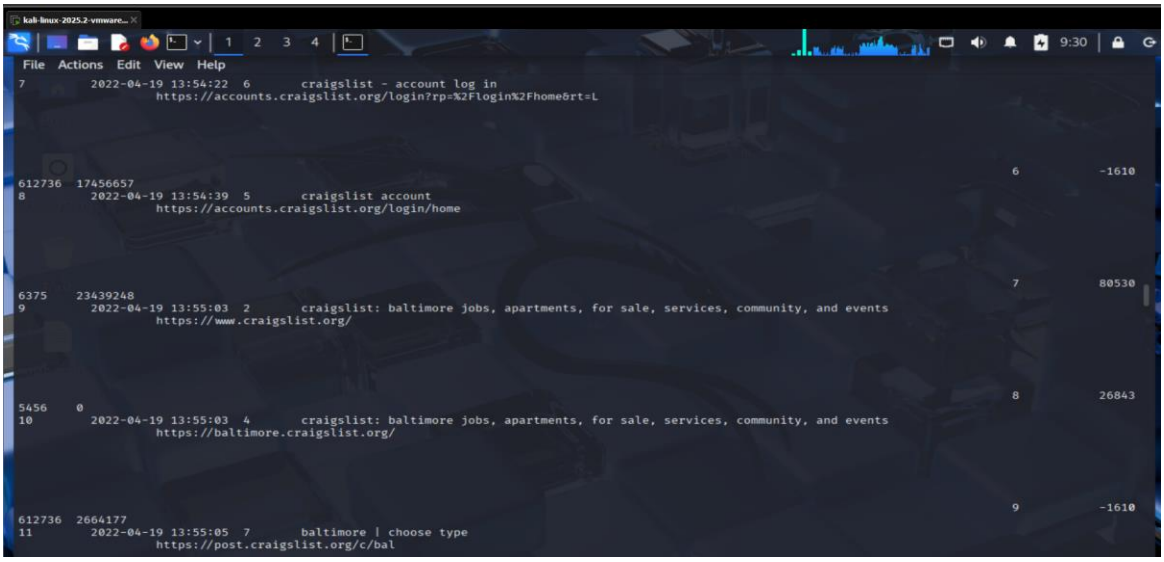
posting, adding a map, previewing the posting, and managing it. At 13:58:31 UTC, a Craigslist page titled “cheaper than Rx supplements - health and beauty - by owner -...” was recorded.

The browser history subsequently records Gmail-related pages, including a message titled “cheaper than Rx supplements - order” at 14:04:12 UTC and further navigation involving that message at 14:04:35 and 14:04:46 UTC. At 14:56:47 UTC, another Gmail page title included a Bitcoin transaction identifier. At 14:58:59 UTC, a Google search for “blockchain explorer” was recorded, followed by a Blockchain.com transaction page at 14:59:06 UTC and a Bitcoin address page at 14:59:21 UTC.

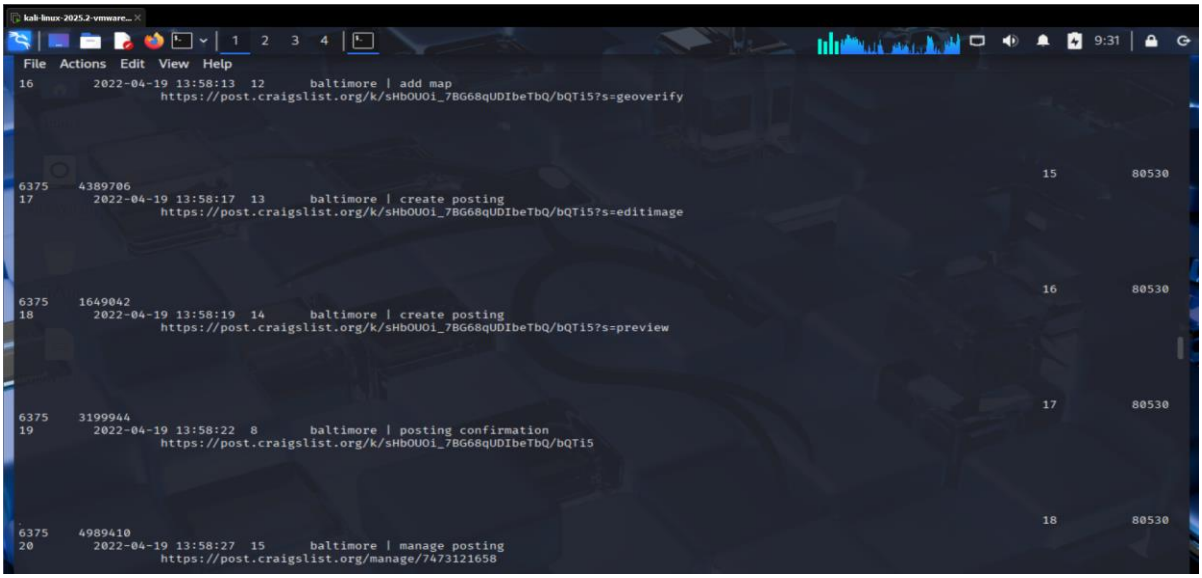
Later records show a Kraken sign-in page at 14:59:47 UTC, a device-approval page at 15:00:21 UTC, a funding overview page at 15:01:21 UTC, and a ledger page at 15:02:05 UTC. Gmail-related pages containing the same transaction identifier in their titles were recorded again between 15:02:32 and 15:02:56 UTC.



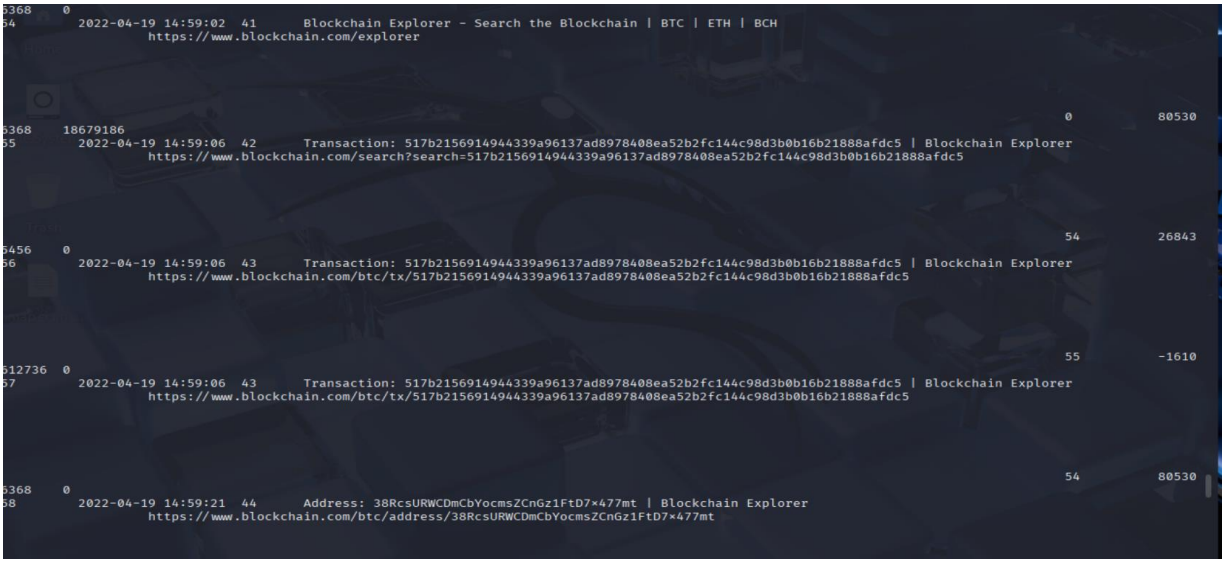

File Actions Edit View Help
6373 0 0 80530
2 2022-04-19 13:54:19 1 craigslist - Google Search
https://www.google.com/search?q=craigslist&rlz=1C1ONGR_enUS989US989&oq=craigslist&aqs=chrome..69i57j46i131i199i433i465i512j0i402l2j0i512l3j0i131i433l2j0i512.2595j0j15&sourceid=chrome&ie=UTF-8
6368 727413 0 80530
3 2022-04-19 13:54:19 2 craigslist: baltimore jobs, apartments, for sale, services, community, and events
https://www.craigslist.org/
5456 0 2 26843
4 2022-04-19 13:54:19 3 craigslist: baltimore jobs, apartments, for sale, services, community, and events
https://geo.craigslist.org/
483648 0 3 -2147
5 2022-04-19 13:54:19 4 craigslist: baltimore jobs, apartments, for sale, services, community, and events
https://baltimore.craigslist.org/
612736 2465460 4 -1610
6 2022-04-19 13:54:22 5 craigslist account
https://accounts.craigslist.org/login/home

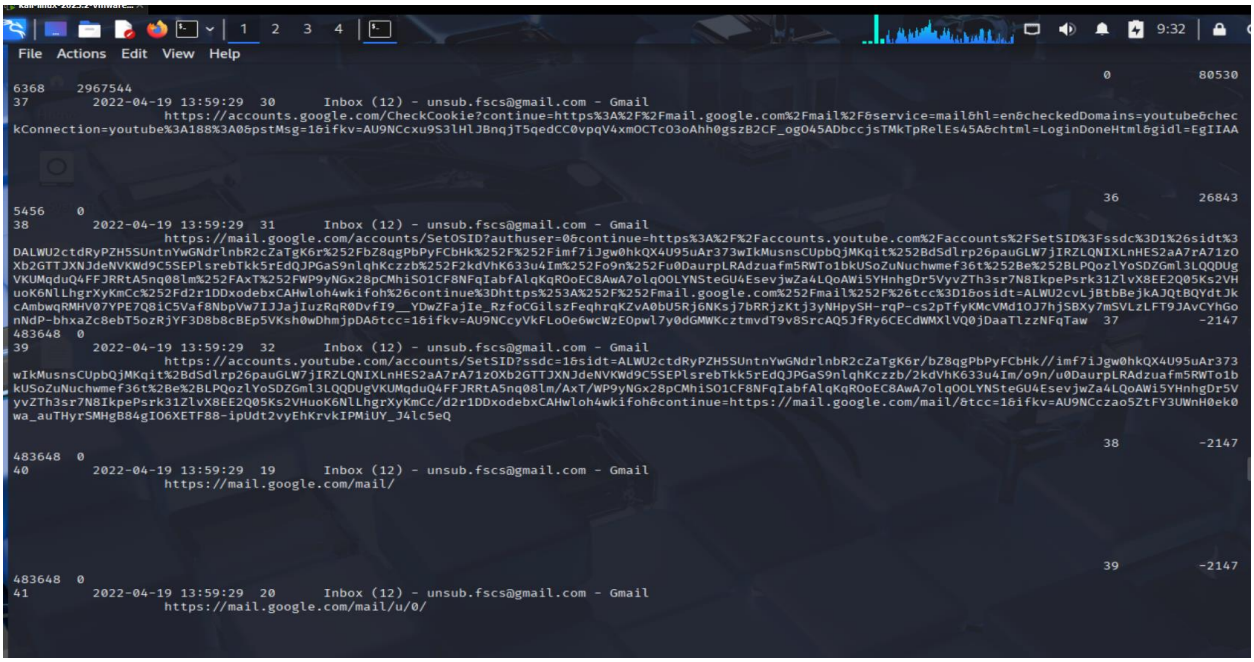

kali-linux-2025.2-vmware...
1 2 3 4
9:30
File Actions Edit View Help
7 2022-04-19 13:54:22 6 craigslist - account log in
https://accounts.craigslist.org/login?rp=%2Flogin%2Fhome&rt=L
612736 17456657 6 -1610
8 2022-04-19 13:54:39 5 craigslist account
https://accounts.craigslist.org/login/home
6375 23439248 7 80530
9 2022-04-19 13:55:03 2 craigslist: baltimore jobs, apartments, for sale, services, community, and events
https://www.craigslist.org/
5456 0 8 26843
10 2022-04-19 13:55:03 4 craigslist: baltimore jobs, apartments, for sale, services, community, and events
https://baltimore.craigslist.org/
612736 2664177 9 -1610
11 2022-04-19 13:55:05 7 baltimore | choose type
https://post.craigslist.org/c/bal

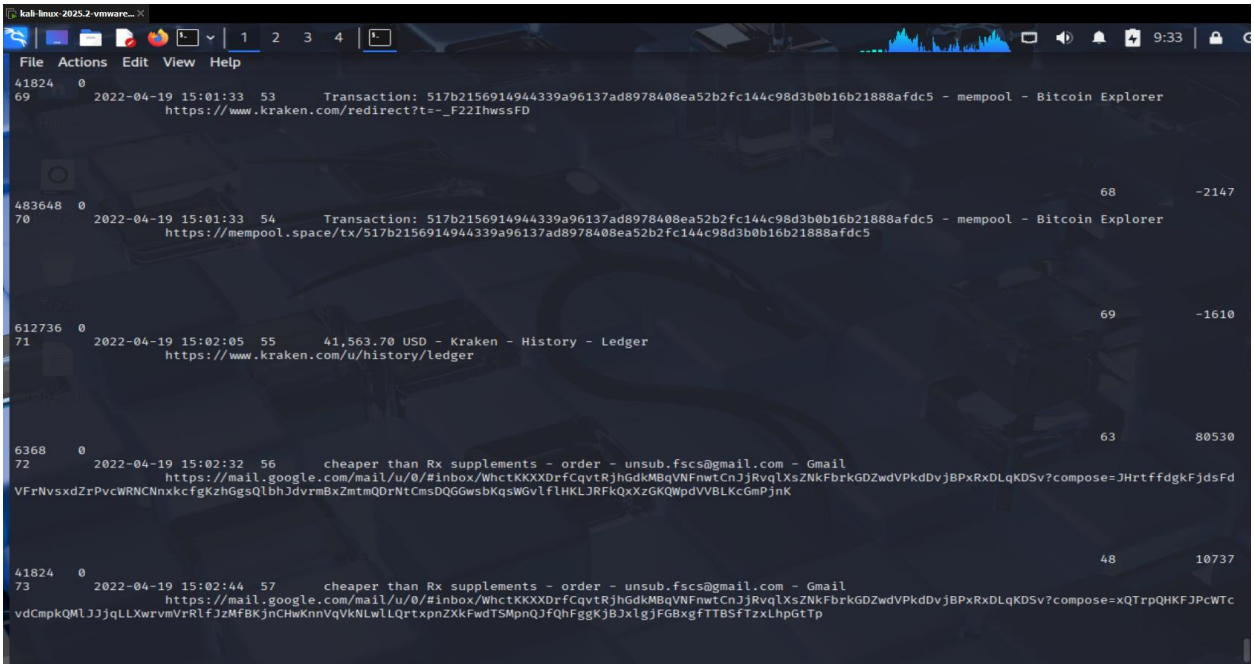

kali-linux-2025.2-vmware...
1 2 3 4
9:30
File Actions Edit View Help
11 2022-04-19 13:55:05 7 baltimore | choose type
https://post.craigslist.org/c/bal
5456 0 10 26843
12 2022-04-19 13:55:05 8 baltimore | posting confirmation
https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5
483648 0 11 -2147
13 2022-04-19 13:55:05 9 baltimore | choose type
https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=type
612736 4273081 12 -1610
14 2022-04-19 13:55:10 10 baltimore | choose category
https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=cat
6375 5599315 13 80530
15 2022-04-19 13:55:15 11 baltimore | create posting
https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=edit


kali-linux-2025.2-vmware...
1 2 3 4
9:31
File Actions Edit View Help
16 2022-04-19 13:58:13 12 baltimore | add map
https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=geoverify
15 80530
6375 4389706
17 2022-04-19 13:58:17 13 baltimore | create posting
https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=editimage
16 80530
6375 1649042
18 2022-04-19 13:58:19 14 baltimore | create posting
https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=preview
17 80530
6375 3199944
19 2022-04-19 13:58:22 8 baltimore | posting confirmation
https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5
18 80530
6375 4989410
20 2022-04-19 13:58:27 15 baltimore | manage posting
https://post.craigslist.org/manage/7473121658


File Actions Edit View Help
16 2022-04-19 13:58:13 12 baltimore | add map
https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=geoverify
15 80530
6375 4389706
17 2022-04-19 13:58:17 13 baltimore | create posting
https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=editimage
16 80530
6375 1649042
18 2022-04-19 13:58:19 14 baltimore | create posting
https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=preview
17 80530
6375 3199944
19 2022-04-19 13:58:22 8 baltimore | posting confirmation
https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5
18 80530
6375 4989410
20 2022-04-19 13:58:27 15 baltimore | manage posting
https://post.craigslist.org/manage/7473121658


6368 0
54 2022-04-19 14:59:02 41 Blockchain Explorer - Search the Blockchain | BTC | ETH | BCH
https://www.blockchain.com/explorer
0 80530
6368 18679186
55 2022-04-19 14:59:06 42 Transaction: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 | Blockchain Explorer
https://www.blockchain.com/search?search=517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5
54 26843
6456 0
56 2022-04-19 14:59:06 43 Transaction: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 | Blockchain Explorer
https://www.blockchain.com/btc/tx/517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5
55 -1610
512736 0
57 2022-04-19 14:59:06 43 Transaction: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 | Blockchain Explorer
https://www.blockchain.com/btc/tx/517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5
54 80530
6368 0
58 2022-04-19 14:59:21 44 Address: 38RcsURWCDmCbYocmsZCnGz1FtD7×477mt | Blockchain Explorer
https://www.blockchain.com/btc/address/38RcsURWCDmCbYocmsZCnGz1FtD7×477mt





A transaction identifier appearing across multiple page titles and URLs provides a useful correlation point for the investigation. However, browser history alone cannot establish whether a transaction was initiated or completed, whether the displayed account belonged to a particular individual, or whether the browser operator authored the associated email. The available records establish a sequence of stored browser visits; further examination of the download metadata and other relevant database records is required before drawing conclusions.

Selected events from the preliminary browsing timeline

UTC date and time	Recorded activity	Evidential significance
19 Apr 2022, 13:54:18	Google search for “craigslist”	Initial recorded event
19 Apr 2022, 13:55–13:58	Craigslist posting workflow pages	Indicates navigation through posting-related pages
19 Apr 2022, 13:58:31	Page title referencing “cheaper than Rx supplements”	Identifies a potentially relevant listing
19 Apr 2022, 14:04:12	Gmail page title referencing the listing and “order”	Potentially relevant email record
19 Apr 2022, 14:56:47	Gmail page title containing a transaction identifier	Correlation lead
19 Apr 2022, 14:58:59–14:59:21	Blockchain search, transaction page, and address page	Related blockchain browsing records
19 Apr 2022, 14:59:47–15:02:05	Kraken sign-in, device approval, funding, and ledger pages	Recorded exchange-related navigation
19 Apr 2022, 15:02:32–15:02:56	Gmail pages associated with the same transaction identifier	Further correlation with the earlier email records

Examination of the Recorded Download

The downloads table contains one download record. The recorded start time was 19 April 2022 at 14:56:58 UTC, and the end time was 14:57:10 UTC, giving a duration of approximately 12 seconds. Chrome recorded 33,844 bytes received out of 33,844 bytes total, with both the MIME type and original MIME type listed as image/png. The opened

field is 0, meaning the database does not record the downloaded item as opened through Chrome's download record.



The fields state, danger_type, and interrupt_reason contain numeric codes. We should not assign meanings to these values without checking the relevant Chrome version's definitions. Although the matching received and total byte counts are consistent with the recorded download completing, we should verify the associated download URL and any available local file before making a stronger conclusion about the file's identity or integrity.

Correlation of the Download with Browsing Activity

The downloads_url_chains table associates download record 1 with the URL https://i.imgur.com/dTgrkP7.png. This identifies the recorded download as a PNG image hosted on Imgur. The download was recorded between 14:56:58 and 14:57:10 UTC on 19 April 2022.



This finding correlates with the browsing timeline, which records a visit to https://imgur.com/dTgrkP7 at approximately 14:56:52 UTC, about six seconds before the download began. The close timing and matching image identifier support the interpretation that the browser visit and download are related. However, the URL and metadata alone do not establish what the image depicts or whether it constitutes evidence of a payment.

Correlation of the Image with the Email Record

The visit timeline shows a Gmail page titled “cheaper than Rx supplements - txid: [transaction identifier] - Gmail” at 14:56:47 UTC on 19 April 2022. Five seconds later, at 14:56:52 UTC, the browser recorded visits associated with Imgur image identifier dTgrkP7. The download began at 14:56:58 UTC and ended at 14:57:10 UTC.

The close sequence provides a useful investigative lead: the Gmail page, Imgur visit, and PNG download occurred within approximately 23 seconds. The shared timing supports further examination of whether the image relates to the email or the transaction identifier. However, browser history alone does not establish that the email contained the image, that the image depicts a payment, or that a cryptocurrency transaction was completed.



Correlation of the Download with Nearby Browser Activity

Key component	Forensic finding
Download record ID	1
Download URL	https://i.imgur.com/dTgrkP7.png

Recorded download start	19 April 2022, 14:56:58 UTC
Recorded download end	19 April 2022, 14:57:10 UTC
Downloaded file type	PNG image
Immediately preceding relevant visit	Imgur page at 14:56:52 UTC
Earlier related browser record	Gmail page at 14:56:47 UTC, with a title containing a transaction identifier
Correlation	The Imgur page visit occurred approximately six seconds before the download began
Evidential limitation	The browser records establish a temporal and URL association, but do not establish the image's contents or prove that it represents a payment

The recorded sequence shows a Gmail page with a transaction identifier in its title, followed by an Imgur page visit and a PNG image download. This sequence provides a useful lead for further examination of the image and its relationship to the other browser records. However, temporal proximity does not by itself prove that the Gmail message led to the download or that the image is payment evidence.

Download Path and Potential Payment Evidence

The Chrome downloads table records the downloaded image as proof_of_payment.png, with both current_path and target_path set to C:\Users\FSCS_User\Desktop\proof_of_payment.png. The record identifies the file as a PNG image and reports 33,844 bytes received out of 33,844 bytes total. This filename provides a specific lead for examining potential payment evidence in the context of the nearby Gmail, Imgur, and cryptocurrency-related browsing activity. However, the path is

a Windows location recorded in the database, and its presence does not confirm that the file itself was included in the acquired evidence or remains available at that location.



Downloaded File Metadata

Component	Recorded value
Download ID	1
Filename	proof_of_payment.png
Recorded target location	C:\Users\FSCS_User\Desktop\proof_of_payment.png
MIME type	image/png
Bytes received	33,844
Total bytes recorded	33,844
Download URL	https://i.imgur.com/dTgrkP7.png
Evidential limitation	The database metadata does not establish the image's contents or authenticity

Examination of Cryptocurrency-Related Browser Records

The URL records show a sequence of visits to a blockchain explorer, a Bitcoin address page, Kraken sign-in and account-related pages, and a mempool.space transaction page on 19 April 2022. The transaction identifier 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 appears in the recorded titles and URLs for both Blockchain.com and mempool.space. This repeated identifier provides a useful correlation point for the investigation.



The records also show a Kraken device-approval page at 15:00:21 UTC, followed by account-related pages. Titles containing dollar amounts were recorded at 15:01:21 and 15:02:05 UTC. These are browser-stored page titles, not independently verified account balances or proof that a transaction was made by the person associated with the browser.

Key Cryptocurrency-Related Browser Records

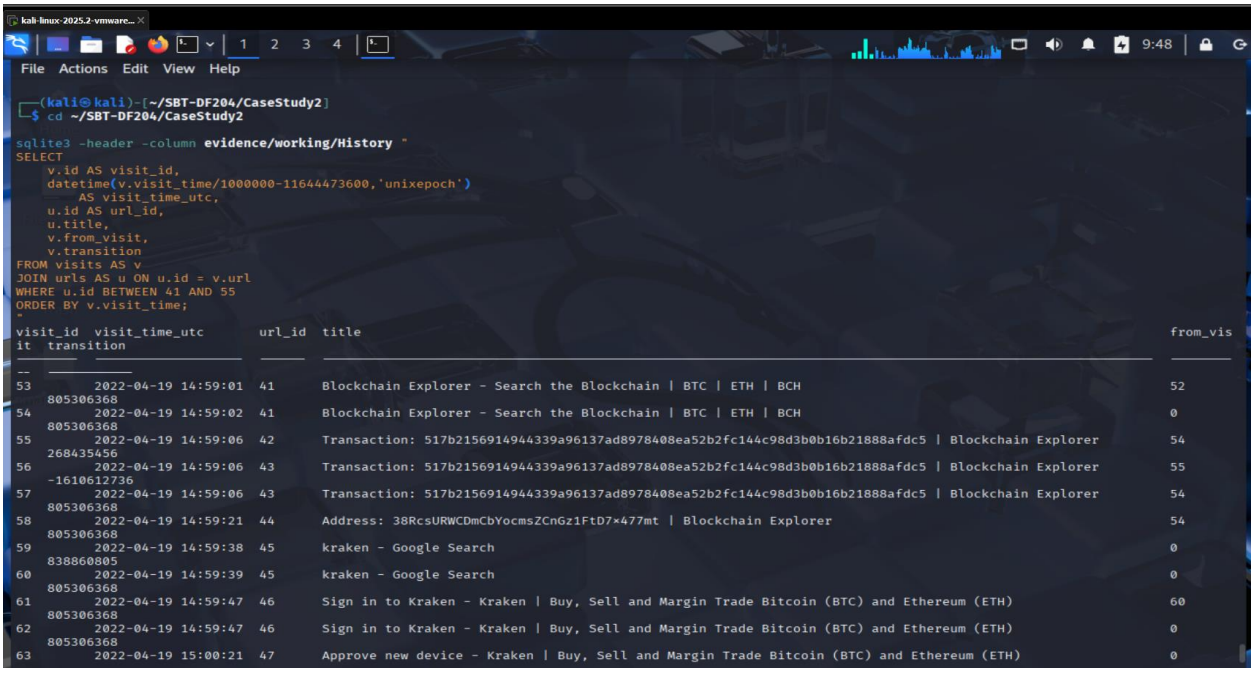
Time (UTC)	Recorded page or activity	Forensic significance
14:59:02	Blockchain.com explorer	Browser activity involving a blockchain explorer
14:59:06	Transaction page containing the recorded transaction ID	Provides a transaction identifier for correlation
14:59:21	Bitcoin address page	Indicates a visit to a specific Bitcoin address page
14:59:47	Kraken sign-in page	Records access to a cryptocurrency exchange sign-in page
15:00:21	Kraken device-approval page	Indicates a device-approval workflow was displayed
15:01:11 – 15:01:21	Kraken trade, instant, and funding pages	Records visits to exchange account-related pages

15:01:31 – 15:01:33	Kraken redirect and mempool.space transaction page	The same transaction identifier appears in the recorded navigation
15:02:05	Kraken ledger-history page	Records a visit to a ledger-history page
Correlation identifier	517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5	Repeated across the recorded transaction-related pages

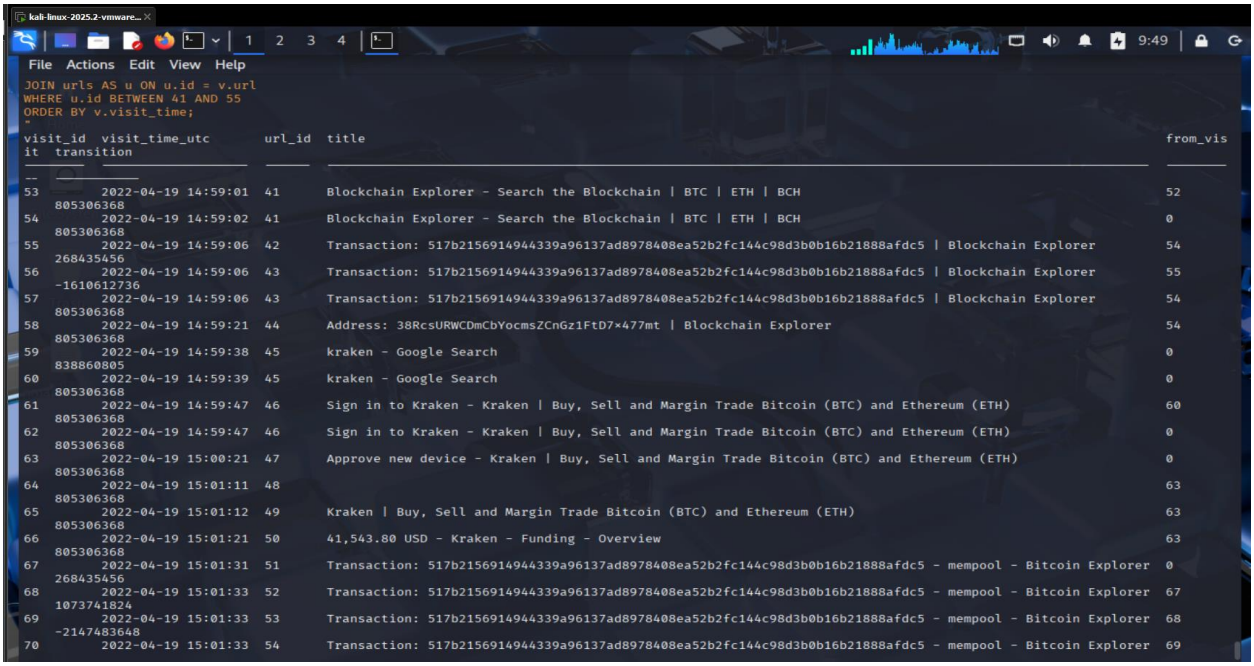
Interpretation: These records support the conclusion that the browser accessed cryptocurrency-related pages during the period under examination. They do not, by themselves, prove that a transaction was initiated, completed, or controlled by a particular person

Reconstruction of Cryptocurrency-Related Navigation

The visit-level records provide more detail than the URL summary. At 14:59:01–14:59:06 UTC, the browser recorded visits to Blockchain.com and a transaction page containing the same transaction identifier later observed on mempool.space. A Bitcoin address page was recorded at 14:59:21 UTC, followed by a Google search for Kraken at 14:59:38 UTC and a Kraken sign-in page at 14:59:47 UTC.



The records then show a device-approval page at 15:00:21 UTC, followed by Kraken account-related pages and a transaction page on mempool.space. Several entries occur within the same second and share parent visit identifiers, indicating related navigation records rather than necessarily distinct user actions. The from_visit field helps explore these relationships, but the numeric transition values should be decoded using the appropriate Chromium definitions before drawing conclusions about the exact navigation method.



Selected Cryptocurrency-Related Visit Events

Visit time (UTC)	Recorded event	Interpretation
14:59:01–14:59:02	Blockchain.com explorer	Initial explorer activity
14:59:06	Transaction page on Blockchain.com	Transaction identifier recorded in the page title
14:59:21	Bitcoin address page	Related address examined in the browser

14:59:38–14:59:47	Google search followed by Kraken sign-in	Transition from a search to an exchange sign-in page
15:00:21	Kraken device-approval page	Device-approval page recorded
15:01:11–15:01:21	Kraken account-related pages	Trade, instant, and funding pages recorded
15:01:31–15:01:33	Kraken redirects and mempool.space transaction page	Repeated navigation records involving the same transaction identifier
15:02:05	Kraken ledger-history page	Ledger-history page recorded

These are browser-history observations, not proof of account ownership, successful authentication, or a completed cryptocurrency transaction.

Examination of Download Referrer and Tab Context

The download record contains a referrer value of https://imgur.com/ and a tab URL of https://imgur.com/dTgrkP7. These values are consistent with the earlier browser-history records showing a visit to the Imgur page followed by a download of https://i.imgur.com/dTgrkP7.png. The download's recorded start and end times remain 14:56:58 and 14:57:10 UTC, respectively.



The site_url and http_method fields are empty in this record. Therefore, the database does not provide values for those fields in the examined row. The referrer and tab URL

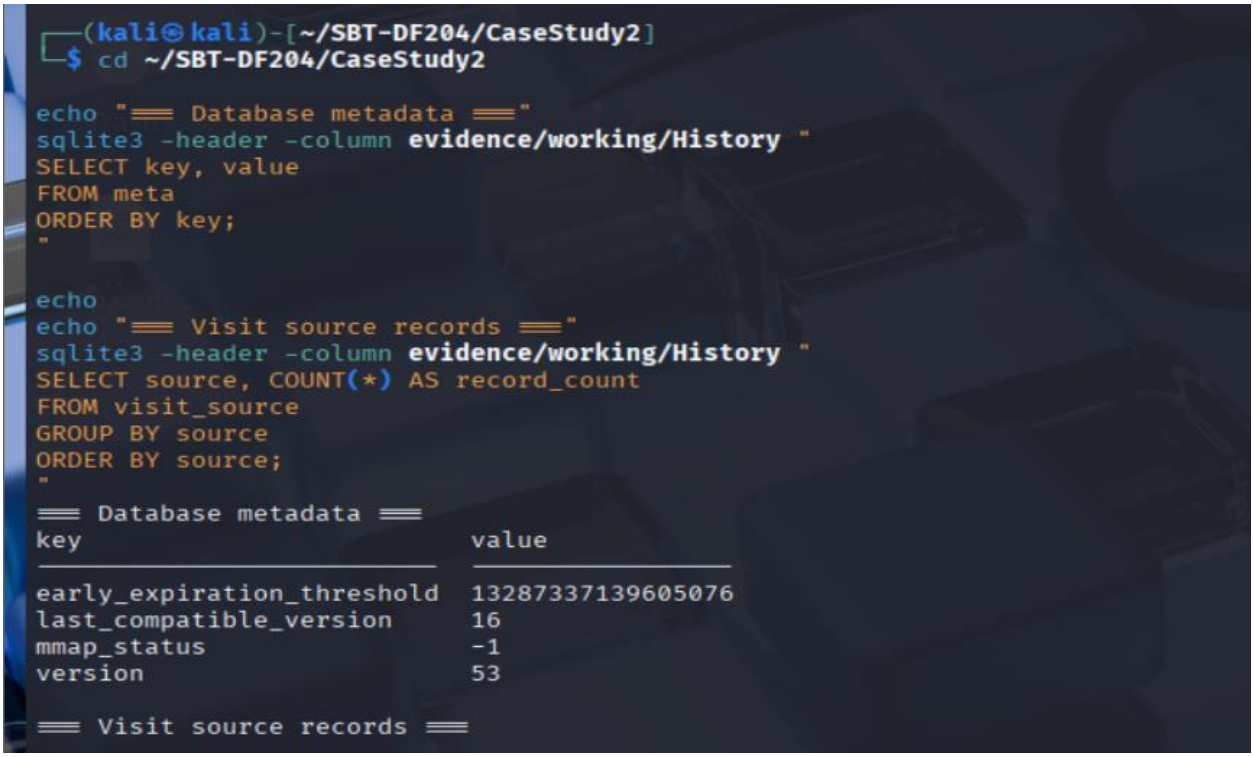
strengthen the association between the Imgur page and the downloaded PNG, but they do not reveal the image's actual contents or prove that it is authentic payment evidence.

Download Context Fields

Field	Recorded value	Significance
Download start	19 April 2022, 14:56:58 UTC	Recorded beginning of download
Download end	19 April 2022, 14:57:10 UTC	Recorded end of download
Referrer	https://imgur.com/	Records an Imgur referrer
Tab URL	https://imgur.com/dTgrkP7	Identifies the associated Imgur page
Site URL	Empty	No value recorded in this field
HTTP method	Empty	No value recorded in this field
MIME type	image/png	Browser identifies the downloaded content as a PNG image

Examination of Database Metadata and Visit Sources

The meta table contains four records: early_expiration_threshold, last_compatible_version, mmap_status, and version. The database records version as 53 and last_compatible_version as 16. These values provide technical context about the stored Chrome database, but should not be interpreted as the Chrome browser's exact version without corroborating evidence.



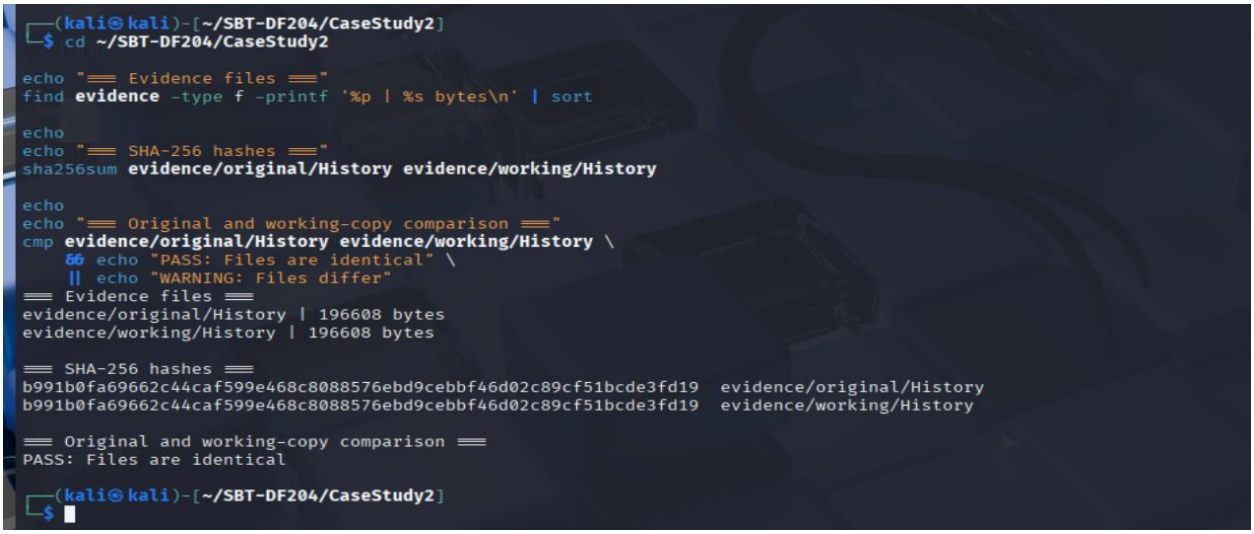
The query against visit_source returned no rows. This means that no visit-source records were available in that table in the examined database. It does not mean that the browsing history is invalid or that the browser activity did not occur; it simply limits what can be established from this particular table.

Database Metadata and Visit-Source Results

Component	Result	Interpretation
Database version	53	Version value recorded in the metadata
Last compatible version	16	Compatibility value stored by the database
Early expiration threshold	13287337139605076	Stored internal threshold; not interpreted independently
Memory-mapped I/O status	-1	Internal database metadata value
Visit-source records	0 rows	No records returned from visit_source

Evidence Integrity Verification

The final integrity check confirmed that the acquired Chrome History database and its working copy are both 196,608 bytes. The SHA-256 hash calculated for each file is identical: b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19. The cmp command also returned a pass result, confirming that the two files are byte-for-byte identical at the time of verification.



This result supports the integrity of the working copy relative to the acquired file. It does not independently establish the provenance or completeness of the original database, so the source and acquisition method should remain documented in the report.

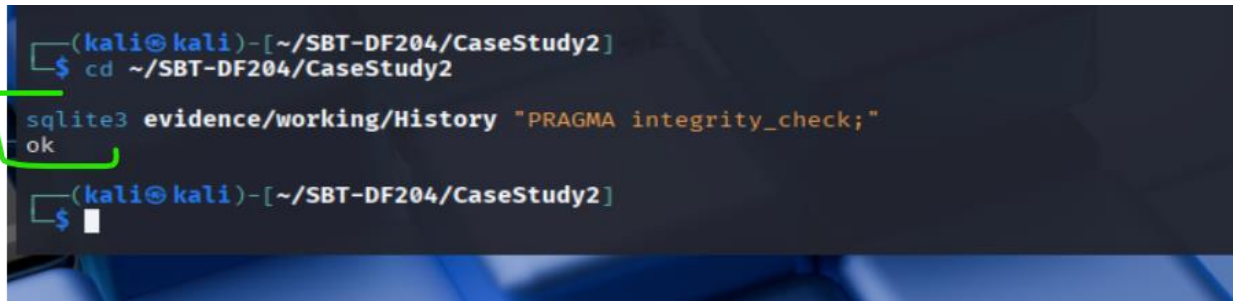
Evidence Integrity Verification Results

Verification item	Result
Original database size	196,608 bytes
Working copy size	196,608 bytes
Original SHA-256	b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19

Working-copy SHA-256	b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19
Hash comparison	Identical
Byte-level comparison	PASS: Files are identical

SQLite Database Structural Integrity Check

The SQLite PRAGMA integrity_check command returned ok when executed against the working copy of the Chrome History database. This indicates that SQLite did not identify structural integrity errors during this check. Together with the matching SHA-256 hashes and successful byte-level comparison, the result supports the use of the working copy for the examination.



However, a successful integrity check does not prove that the database contains every historical browser event, that the source data is authentic, or that the recorded activity can be attributed to a particular individual.

SQLite Structural Integrity Results

Check	**Result**
Database examined	evidence/working/History
Command	PRAGMA integrity_check;
Output	ok

Structural errors reported	None
Limitation	Does not establish provenance, completeness, or user attribution

Summary of Findings

The examination of the Chrome History SQLite database identified 57 URL records, 74 visit records, four recorded search terms, and one download record. The recorded visits span 19 April 2022, from 13:54:18 UTC to 15:02:56 UTC. The search terms were *craigslist*, *gmail*, *blockchain explorer*, and *kraken*. These records provide a basis for reconstructing the browsing activity represented in the database.

The browsing history contains a sequence of Craigslist posting-related pages, Gmail pages, and cryptocurrency-related pages. A Craigslist page title referred to “cheaper than Rx supplements,” while a Gmail page title included the same subject and the word “order.” Later records contain a transaction identifier associated with Blockchain.com and mempool.space, as well as visits to Kraken sign-in, device-approval, funding, and ledger-history pages. These observations indicate related browsing activity, but the available records do not independently establish the identity of the person who performed the actions or prove that a particular transaction was completed.

The database also records a PNG download named proof_of_payment.png, with a recorded Windows destination of C:\Users\FSCS_User\Desktop\proof_of_payment.png. Chrome records the download URL as https://i.imgur.com/dTgrkP7.png, with a recorded download interval of 14:56:58 to 14:57:10 UTC. The download's referrer and tab URL are consistent with a visit to Imgur shortly beforehand. However, the image was not located in the case evidence directories or the searched Downloads directory, so its contents and authenticity could not be independently examined.

Evidential Significance

The strongest findings are the recorded browser events and the correlations between them. The repeated transaction identifier provides a useful link between browser records

for different cryptocurrency-related pages. The download metadata also connects the recorded PNG file with an Imgur page visit. These correlations can guide further investigation and help identify which artefacts should be acquired for a more complete examination.

Nevertheless, Chrome History is a record of browser activity, not a complete record of a person's actions or intentions. Page titles, URLs, visit relationships, and download metadata do not independently prove that the user successfully authenticated to an account, controlled a cryptocurrency wallet, sent or received funds, or created an advertisement for an unlawful purpose. The evidence must therefore be interpreted in conjunction with other independently acquired artefacts.

Limitations

The examination was limited to the supplied Chrome History database. The image referenced by the download record was not available in the searched directories, and no independent examination of its contents was possible. The investigation did not include a forensic image of the original Windows system, browser cache files, email message contents, exchange records, wallet artefacts, or independent blockchain analysis.

The source and relevance of the sample database must also be confirmed against the official assessment instructions. The database's successful SQLite integrity check and matching SHA-256 hashes establish that the working copy matched the acquired file and that SQLite did not identify structural integrity errors. These checks do not independently prove the provenance, completeness, or authenticity of the original source evidence.

Further Examination Recommended

A follow-up examination should seek an authorised forensic acquisition of the source Windows system or the relevant original artefacts. Priority should be given to locating the recorded proof_of_payment.png file, examining its metadata and contents, and comparing its hash with any independently acquired copy. If authorised email evidence is available, the relevant message should be examined with its headers and surrounding correspondence preserved. Any cryptocurrency transaction should be evaluated using independently obtained blockchain records and, where legally authorised and available,

relevant exchange records. These steps would help test the correlations identified in the browser history without treating them as proof of transaction completion or individual attribution.

Conclusion

The examination reconstructed a sequence of browser activity involving Craigslist, Gmail, an Imgur imagConclusione download, blockchain explorers, and cryptocurrency exchange pages on 19 April 2022. The Chrome History database records a PNG download named proof_of_payment.png and several cryptocurrency-related visits that share a transaction identifier. These findings provide investigative leads relevant to the assessment scenario.

The available evidence is insufficient to verify the image's contents, establish that a cryptocurrency transaction was completed, or attribute the recorded activity to a specific individual. The conclusions are therefore limited to the browser records examined. Further examination of the original system, the referenced image, relevant email artefacts, and independently obtained transaction records would be required to reach stronger conclusions.